

VLSI Design

BRANCH: ELECTRONICS (ECE)

COURSE CODE: EC402

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for VLSI Design (EC402) under the Electronics (ECE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: MOS Transistor Theory

nMOS, pMOS enhancement transistors, threshold voltage, MOS equations, channel length modulation, body effect. CMOS fabrication steps: p-well, n-well, twin-tub process.

Unit 2: CMOS Inverter & Logic

CMOS Inverter DC characteristics, beta ratio, noise margins. Delay analysis, fall and rise times. Design of NAND, NOR, and Complex CMOS logic gates.

Unit 3: Sequential & Dynamic CMOS

Transmission gates, pass-transistor logic. Dynamic logic basics, domino logic, charge sharing. CMOS memory: 6T SRAM cell, DRAM.

Unit 4: Layout & Testing

CMOS layout design rules (lambda-based). Stick diagrams. Design for Testability (DFT): Scan path design, Built-In Self-Test (BIST), Boundary scan.

*MOS Drain Current (Sat): $I_d = (\beta / 2) * (V_{gs} - V_t)^2$. CMOS static power dissipation is approximately zero.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



